# Technology Stack

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| **Date** | 8 July 2026 |
| **Team ID** | SWTID-2026-5447 |
| **Project Name** | Credit Card Approval Prediction |
| **Maximum Marks** | 2 Marks |

**Define Your Technology Stack:**

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements. Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

**Technology Stack Details:**

| S No | Architecture Component / Layer | Technology Chosen | Justification / Purpose |
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| 1 | **Frontend / Client-Side** | HTML5, CSS3, Bootstrap, JavaScript | To create a responsive and userfriendly web interface for applicants to submit details and view results. Bootstrap ensure mobile compatibility. |

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| 2 | **Backend / Server-Side** | Python, Flask | *Python is best for ML integration. Flask is lightweight and easy to deploy REST APIs for prediction logic.* |
| 3 | **Database / Data Storage** | CSV File/SQLite | Used to store applicant data locally. Since we are not using a full database server, CSV/SQLite is sufficient for this prototype. |
| 4 | **Cloud / Hosting / Deployment** | Localhost/Not Deployed Yet | Application is currently running on local machine using Flask. Cloud deployment can be done in future. |
| 5 | **Version Control & CI/CD** | GitHub, Git | GitHub is used for code versioning and team collaboration. It enables tracking changes and seamless deployment of code updates. |
| 6 | **Third-Party APIs / Other Tools** | Jupyter Notebook, Pandas, Scikit-learn | Jupyter Notebook for model training and analysis. Pandas for data handling. Scikit-learn for ML model building and prediction. |